

MeriTayyari

Notes - Tuba

September 2026

Table of Contents

- 1. Newton's Laws
- 2. Newton's Laws (Full)

Newton's Laws

physics - Grade 9th

$$F = ma$$

Newton's Laws (Full)

physics - Grade 9th

Newton's Second Law of Motion

- **Statement:**

- The rate of change of momentum of an object is directly proportional to the net external **force** applied, and occurs in the direction of that force.

- **Mathematical form (constant mass):**

$$F_{net} = m a$$

- **F** = net external force
- **m** = mass of the body
- **a** = acceleration produced

- **Key points:**

- Force, mass, and acceleration are vectors; direction matters.
- In absence of external force, an object maintains its state of motion (inertia).

- **Applications:**

- Predicting motion of cars, rockets, or any accelerating body.
- Calculating required force to change velocity.